

EC-490 – TIME SERIES ECONOMETRICS

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Lectures and office hours:

- Lectures
 - Location: CAS B36
 - Time: Mon, Wed, and Fri from 1:25 PM until 2:15 PM
- Office Hours
 - Mon: 2:30-4:00PM; Wed: 9:00AM to 10:30AM
 - Office address: 270 Bay State Road, office # 539
- TA: Yinan Zhang
- TA's office hours: Tue: 10:00 AM to 11:30Am
- Textbook: Time Serie Econometrics: Learning Through Replication by John D. Levendis

GRADING And Problem Sets

- **Grading Criteria**

- Exam: 40%

- Research Paper: 40%

- Problems Sets: 20 %

- For more details about exam policies, problem sets, and other policies, check the course syllabus.

SOFTWARE: STATA

- You are required to purchase the econometric software package STATA. It can be ordered online at:

<http://www.stata.com/order/new/edu/gradplans/student-pricing/>

Installing New STATA Commands

- Over this course you will need to download many publicly available economic and financial data. In order to have access to those data, you will need to install the following commands (just type the following on your STATA command windows):

net from <http://researchata.com/stata/203>

net install wbopendata.pkg

net install st0110_1.pkg (if you type this command and are not able to install this package, just search on STATA: st0110_1.pkg and click to install).

Installing New STATA Commands

- Also, the following commands we will be using during this course and you will need to install on STATA:

`net install sim_arma.pkg` (generates data from an ARMA model)

`net install sts15_2.pkg` (KPSS unit root test command)

`net install zandrews.pkg` (Zivot-Andrews unit root test command)

`Net install st0085_2.pkg` (estout package for reporting regression results)

OBS: If you type `net install` and are not able to install the packages, just search the package name on STATA and click to install).

Specifying Time in STATA

- Different than cross-section data, for time series the order of observation matters!!!
- We will need to let STATA know that we are working with a time series data.
- Whenever you upload a time series, you will need to type the following commands:

`generate time = _n` (when you generate time, you can specify the period)

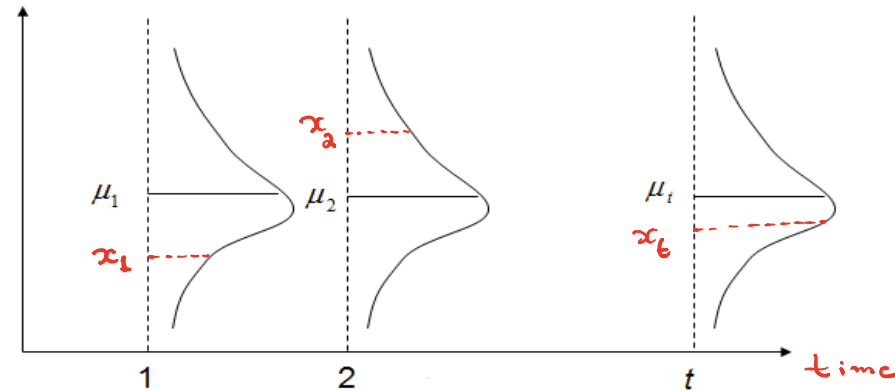
`tsset time`

INTRODUCTION TO TIME SERIES

- **WHAT IS A TIME SERIES?** It is a set of observation generated sequentially in time. (equally spaced time interval; it can be discrete or continuous): $X_t = \{x_1, x_2, x_3, \dots, x_T\}; t = 1, 2, \dots, T$
- **WHAT MAKES TIME-SERIE UNIQUE?** ITS OBSERVATIONS ARE DEPEDENT OR CORRELATED, THE ORDER (LAG) OF THE OBSERVATION MATTERS: x_t and x_{t-1} are correlated
- Main Issue with Time Series: Dependence.
- Depending on assumptions, classical results (based on LLN & CLT) may not be valid.
- **STATISTICAL PROCEDURES THAT RELY ON INDEPEDENCE ASSUMPTION WILL NOT BE APPLICABLE**
 **Different methods will be required.** However, we can use the past to understand and describe the data generating process of the series and forecast the future.

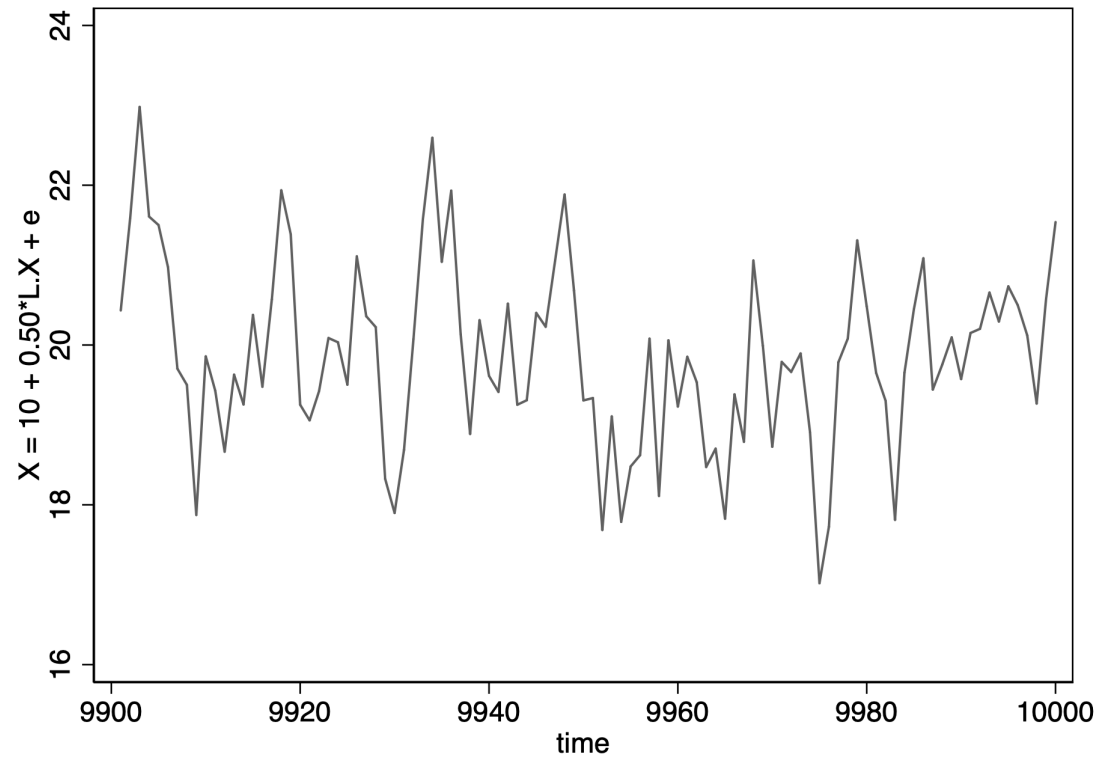
TIME SERIES PROCESS (or a stochastic process)

It is a **set of time indexed random variables** defined on a sample space. It evolves in time according to probabilistic laws. The **observed time series**, $x_1, x_2, \dots, x_t$, is a **particular realization of this process**.



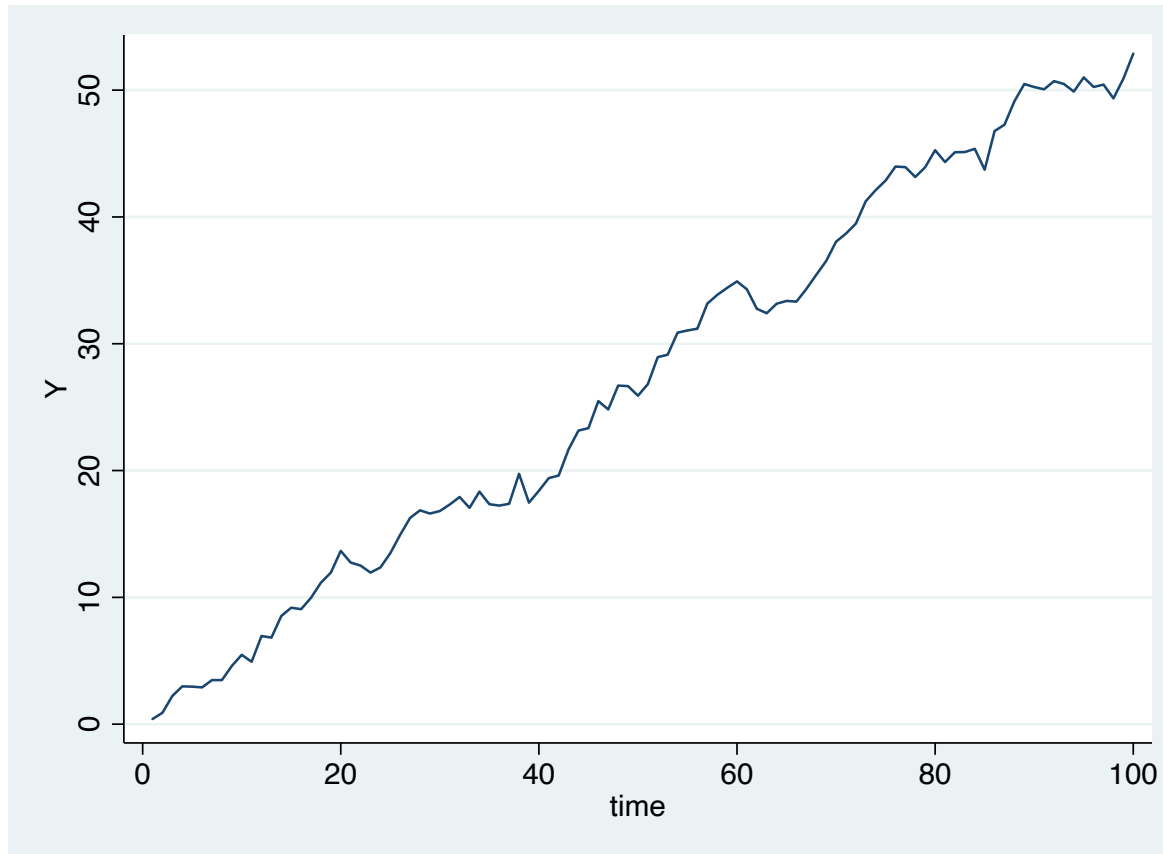
Classes of Stochastic Process

- Stationary process (AR, MA, ARMA)



Classes of Stochastic Process

- Non-stationary process: a random walk with drift: $Y_t = 0.5 + Y_{t-1} + t + e_t$



TIMES SERIES MODELING

- **GOAL:** Develop simple models capable of forecasting, interpreting, and testing hypotheses concerning time series data.
- The time-series methodology we will learn in this course will allow us to uncover the dynamic path of a time series (we can test our models) to get more accurate forecast.
 - For example: day-to-day changes in the price of a stock (we expect to have a mean value of zero); in this case, the price of a stock can be modeling as a random walk process.
- An economic (or financial) time series can contain a stochastic element in its trend, seasonal, and irregular components.
- The time-series methodology involves find the equation of motion (difference equations that express the value of a variable as a function of its own lagged values, time, and other variables) of the process and use the past past to predict the future.

Decomposing the Time Series

- Decomposing the Time into trend, seasonal and irregular components:

$$y_t = \text{trend} + \text{seasonal} + \text{irregular}$$

- **Classical approach:** the trend and seasonal components is viewed as deterministic functions;
- **Modern Approach: they all contain stochastics components**
 - we will first look at the ways how to model the irregular component, and leave seasonal and trend components for later

Trend, Seasonality and Irregular Component

- **trend** is a tendency of the time series to either grow or decline over the long term (a permanent component)
- **seasonality** refers to regular patterns arising in economic activity due to calendar (on quarterly, monthly, day of week basis). Timing of peaks and troughs is predictable with seasonal.
- **Irregular component:** lack of a well-defined pattern, but it is predictable in the short-run (AR, MA, ARMA models)

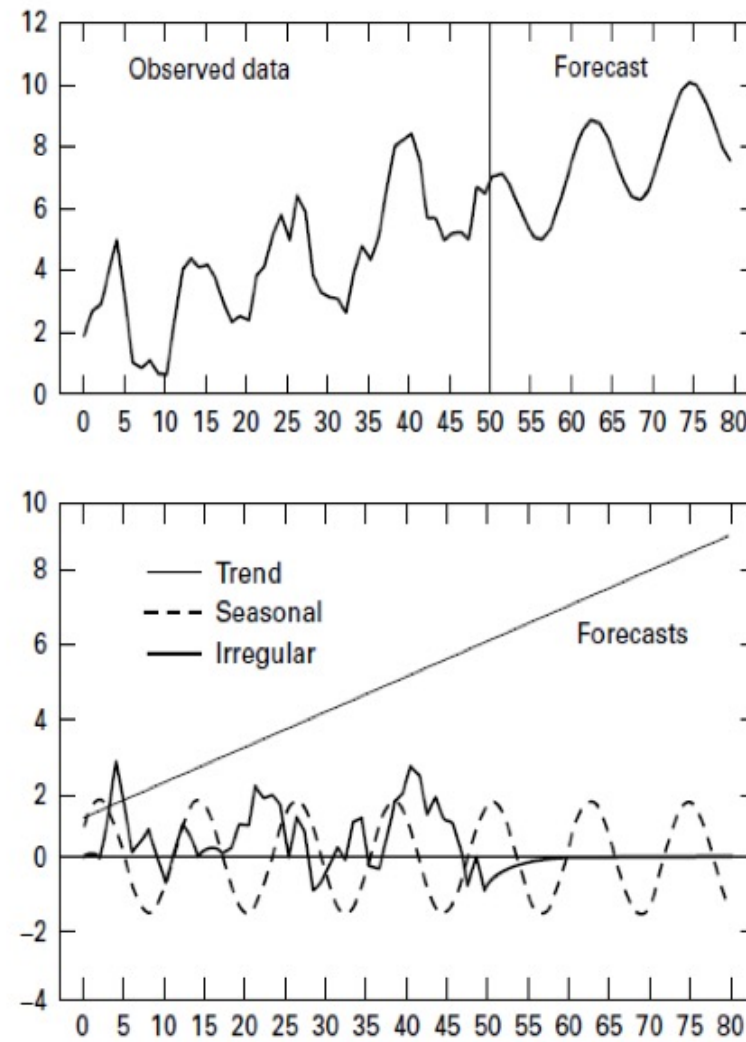
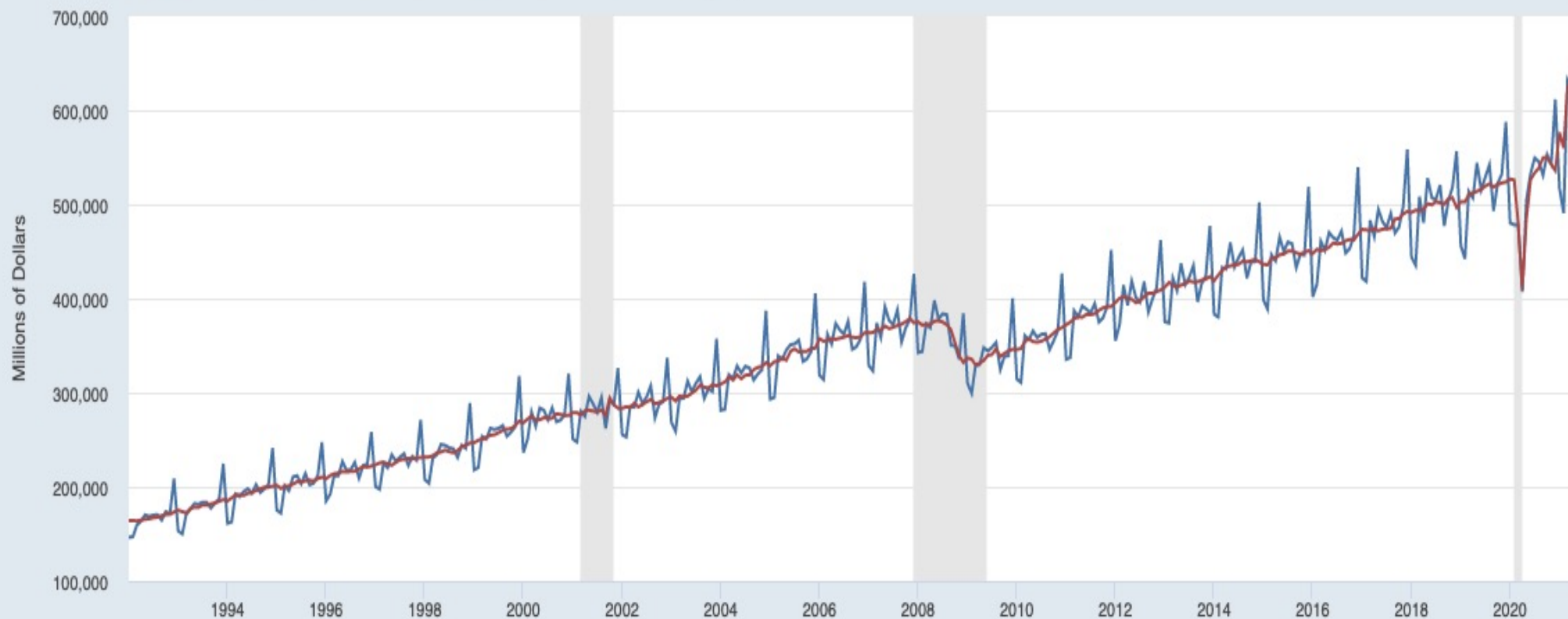


FIGURE 1.1 Hypothetical Time Series

Figure 1.1 From Enders: Applied Econometric Time Series (2003)



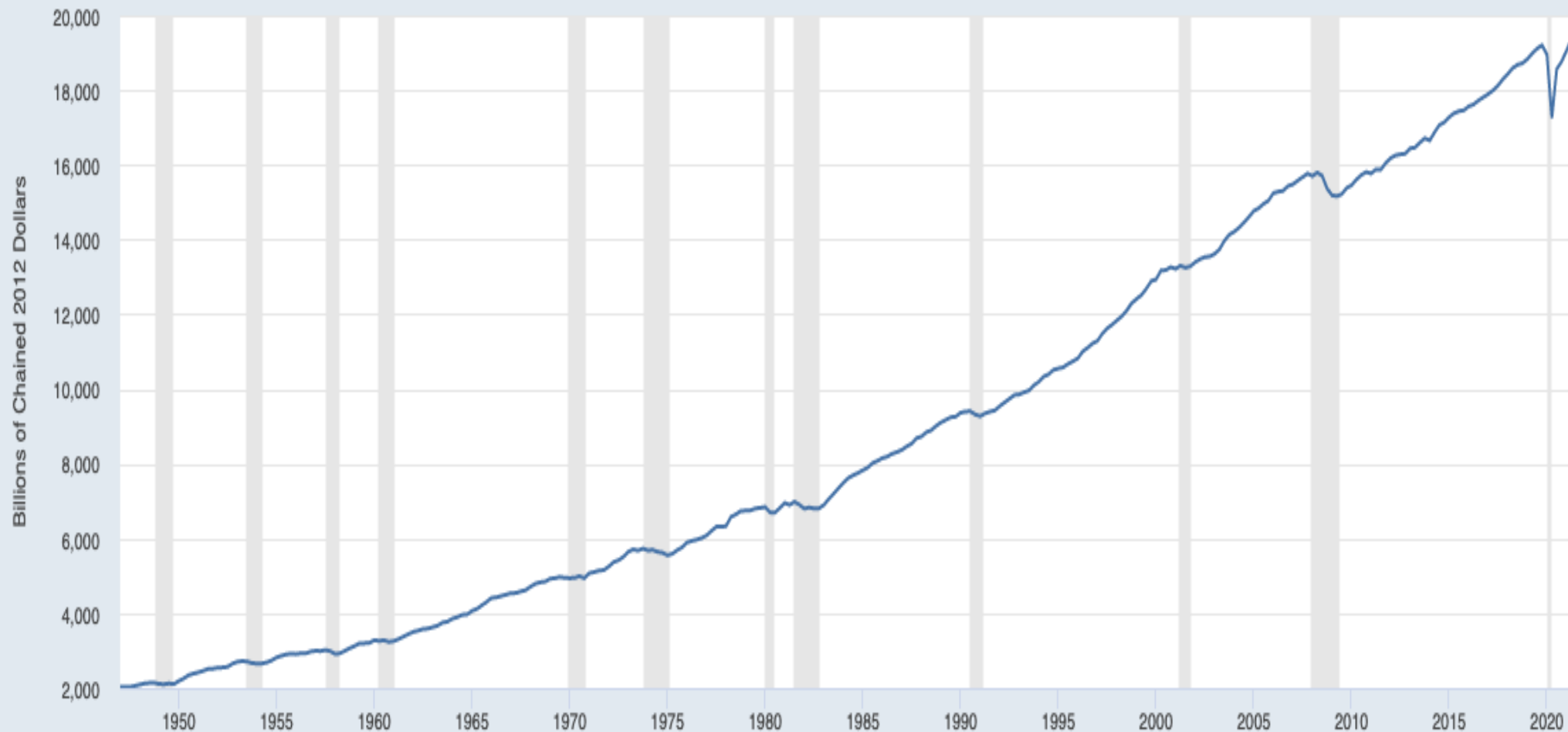
— Advance Retail Sales: Retail Trade and Food Services; Not Seasonally Adjusted
— Advance Retail Sales: Retail Trade and Food Services; Seasonally Adjusted



Shaded areas indicate U.S. recessions.

Source: U.S. Census Bureau

fred.stlouisfed.org



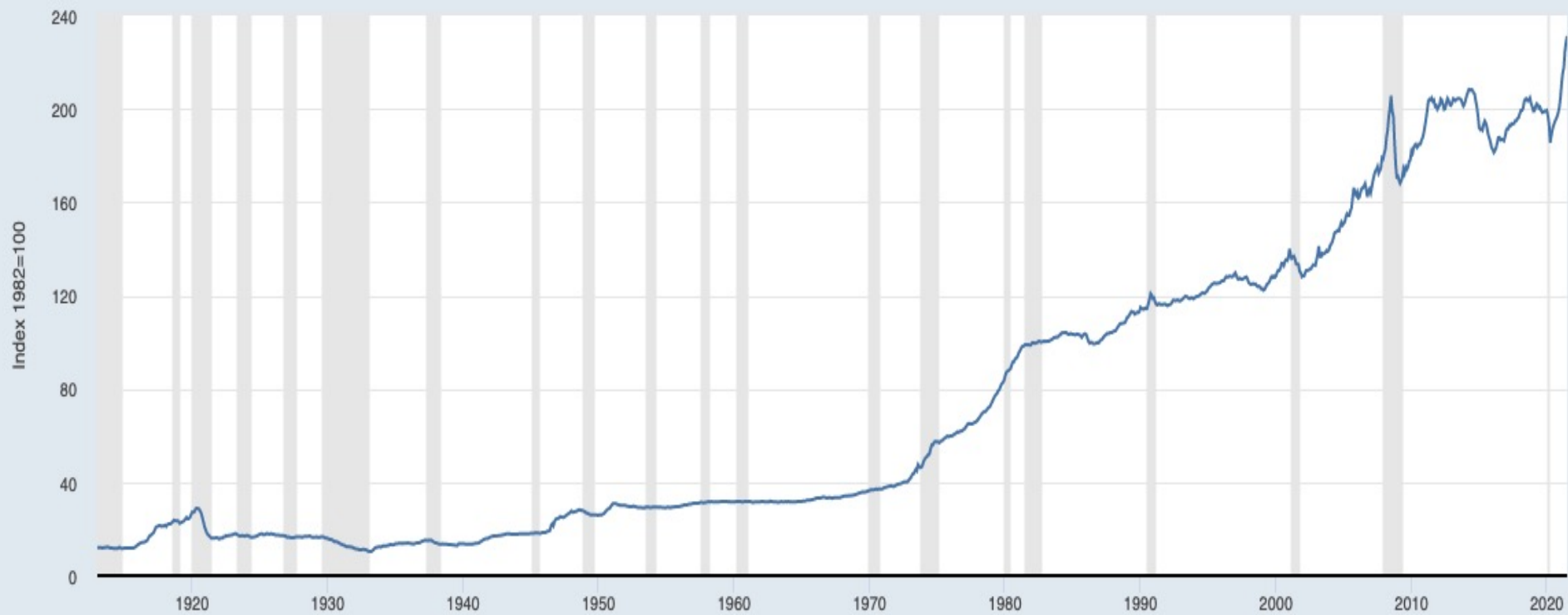
Shaded areas indicate U.S. recessions.

Source: U.S. Bureau of Economic Analysis

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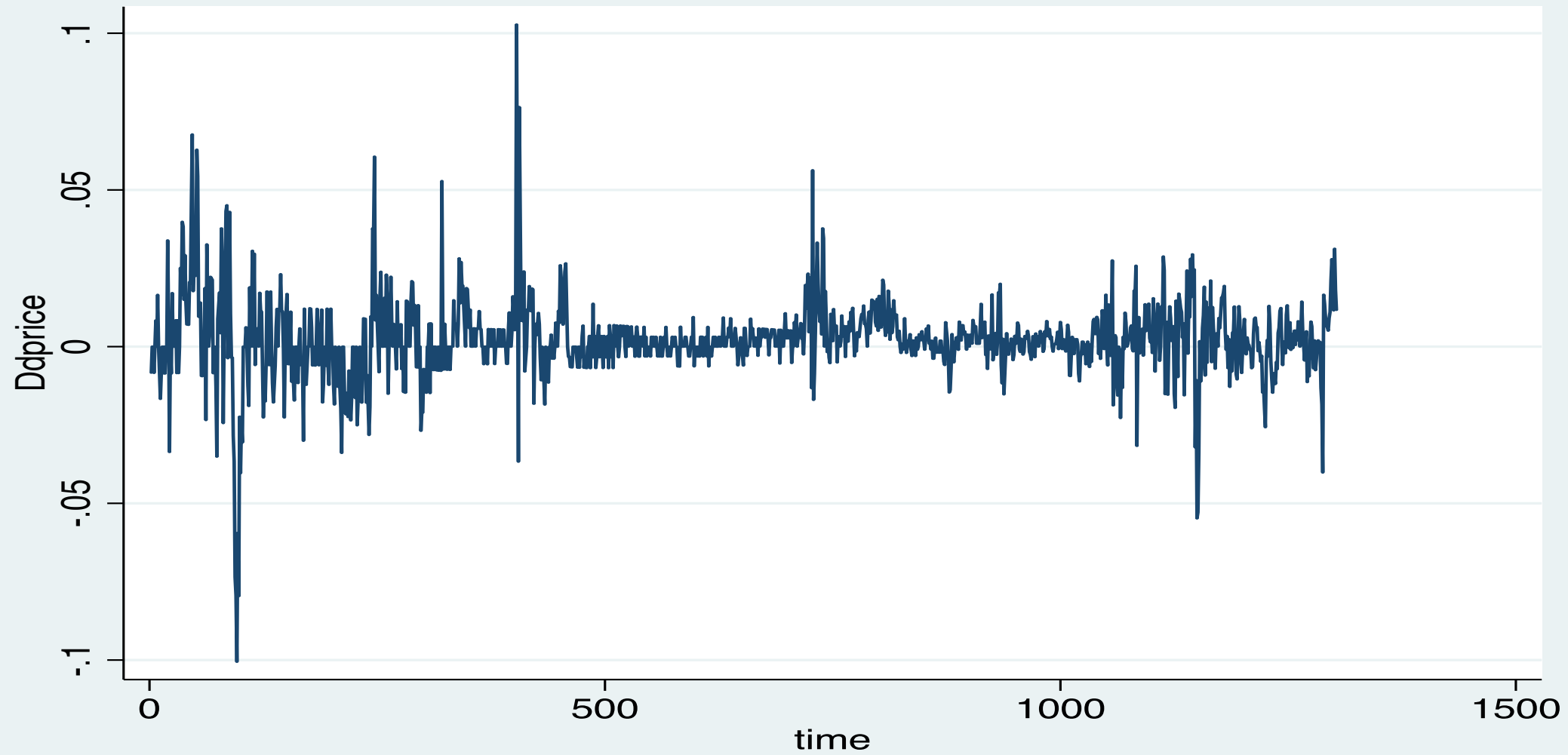
— Producer Price Index by Commodity: All Commodities



Shaded areas indicate U.S. recessions.

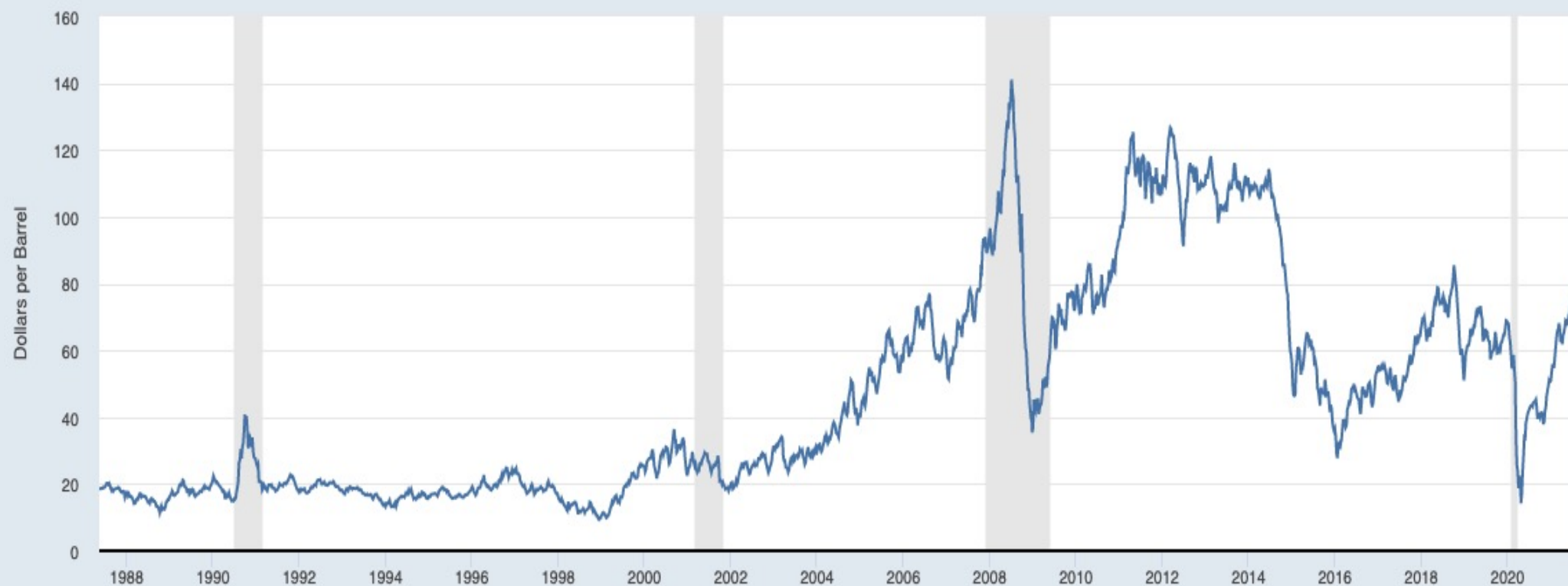
Source: U.S. Bureau of Labor Statistics

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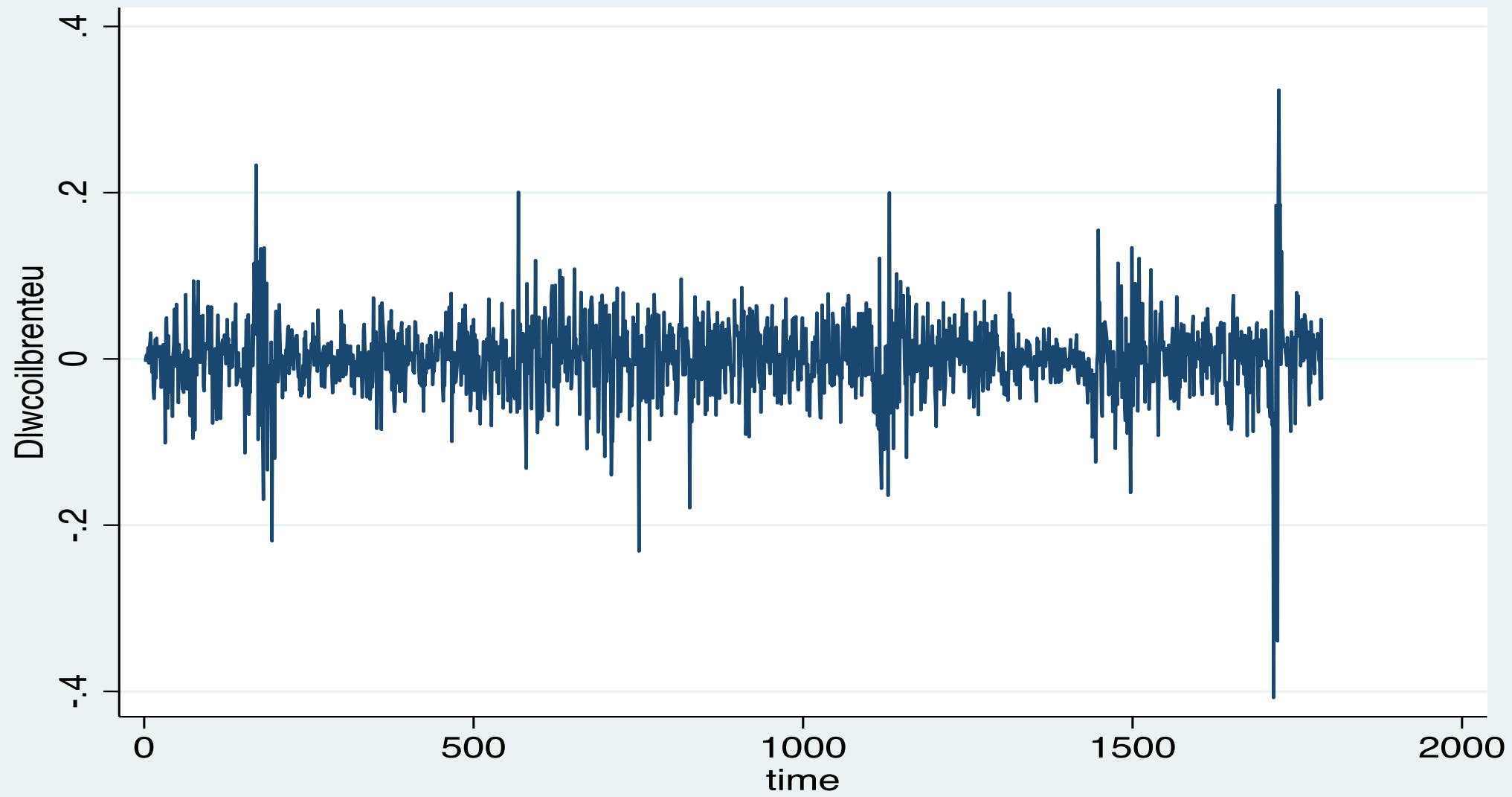
— Crude Oil Prices: Brent - Europe



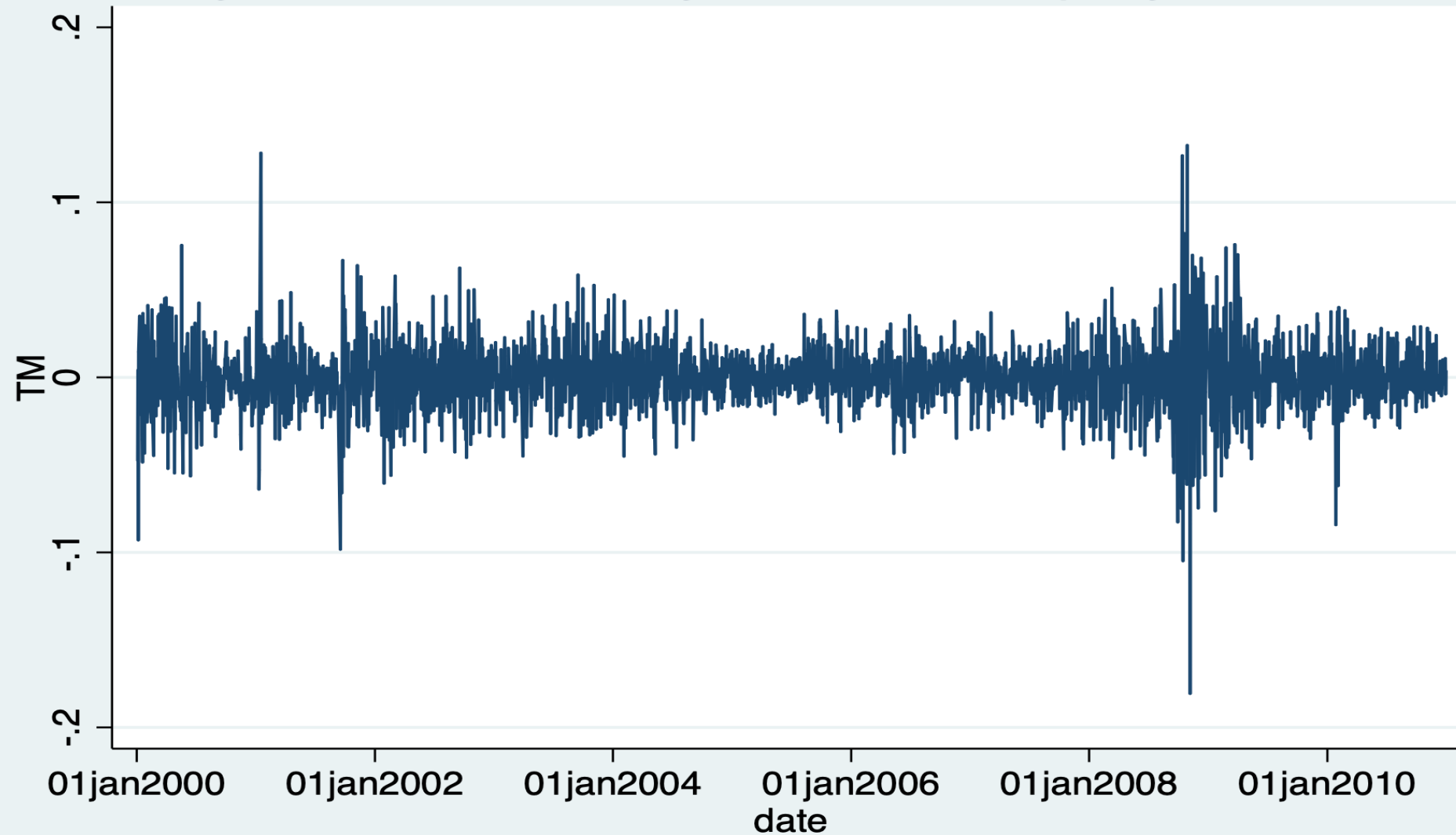
Shaded areas indicate U.S. recessions.

Source: U.S. Energy Information Administration

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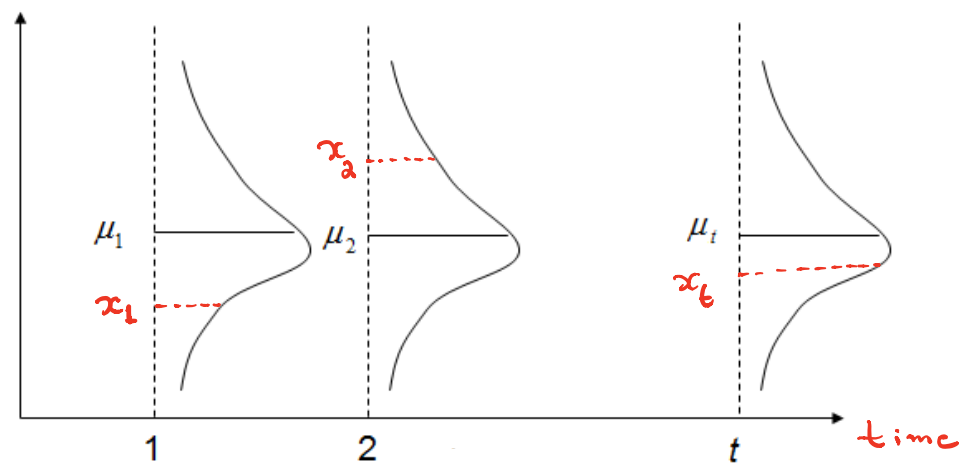
Daily Stock Prices of Toyota Motor Company=2000-2010



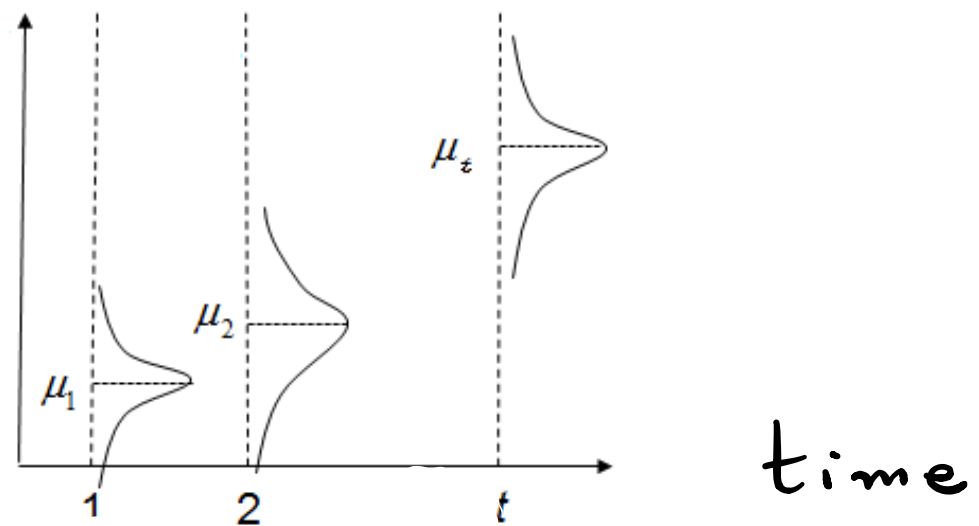
IMPORTANCE OF STATIONARITY

- **Strictly Stationarity:** requires all these moments of the distribution (mean, variance, covariance, kurtosis, etc..) to be independent of time. The whole probability structure must depend only on time differences.
- **Weakly stationarity (or second-order stationary):** constant mean, constant variance and the covariance are independent of time (only depends on time difference). A Gaussian process (normal process) is defined only by its 2 moments, so if it is weakly stationary, then it is strictly stationary.

STATIONARITY PROCESS



NON-STATIONARITY PROCESS



LAG and DIFFERENCE OPERATORS

- The lag operator L is define as:

$$LX_t = X_{t-1}$$

$$L^2X_t = X_{t-2}$$

$$L^pX_t = X_{t-p}$$

- The difference operator is define as:

$$\Delta X_t = X_t - X_{t-1} \quad \text{or} \quad \Delta X_t = (1 - L)X_t$$

$$\Delta^2 (X_t) = \Delta(X_t - X_{t-1}) = (1 - L)^2 X_t$$

STATISTICAL REVIEW

- Probability distribution function of a random variable (RV): assign a probability to each possible outcome of the random variable X.
- Discrete random variables: take on distinct values (integers).
- Continuous random variables: can take any value between two numbers (infinite values).
- For discrete RV:

$$E(X) = \sum x_i Pr(X = x_i) = \mu$$

$$Var(X) = E(X^2) - E(X)E(X) = \sigma^2$$

$$Cov(X, Y) = E(XY) - E(X)E(Y) = \sigma_{x,y}$$

$$Corr(X, Y) = \frac{Cov(X, Y)}{Std(X)Std(Y)} = \rho_{x,y}$$

STATISTICAL REVIEW

- For continuous RV:

$$E(X) = \int xf(x)dx = \mu$$

$$Var(X) = E(X^2) - E(X)E(X) = \int (x_i - E(X))^2 f(x)dx = \sigma^2$$

$$Cov(X, Y) = E(XY) - E(X)E(Y) = \sigma_{x,y}$$

$$Corr(X, Y) = \frac{Cov(X, Y)}{Std(X)Std(Y)} = \rho_{x,y}$$

STATISTICAL REVIEW

- $E(a) = a; \ a = \text{constant}$
- $E(aX) = aE(X)$
- $\text{Var}(aX) = a^2\text{Var}(X)$
- $\text{Cov}(aX, bY) = ab\text{Cov}(X, Y)$
- $\text{Corr}(aX, bY) = \text{Corr}(X, Y) = \text{Corr}(Y, X)$
- $E(a + X) = a + E(X)$
- $\text{Var}(a + X) = \text{Var}(X)$
- $E(aX + bY) = aE(X) + bE(Y)$
- $\text{Var}(aX + bY) = a^2\text{Var}(X) + b^2\text{Var}(Y) + 2ab\text{Cov}(X, Y)$

STATISTICAL REVIEW

- Let e_t be realizations of a variable and each e is drawn independently of each other from the same identical normal distribution (i.i.d):

$$e_t \sim N(0, \sigma^2)$$

$$E(e_t) = 0$$

$$\text{Var}(e_t) = E(e_t^2) = \sigma^2$$

$$\text{Cov}(e_t, e_{j \neq t}) = 0$$

$$\text{Corr}(e_t, e_{j \neq t}) = 0$$

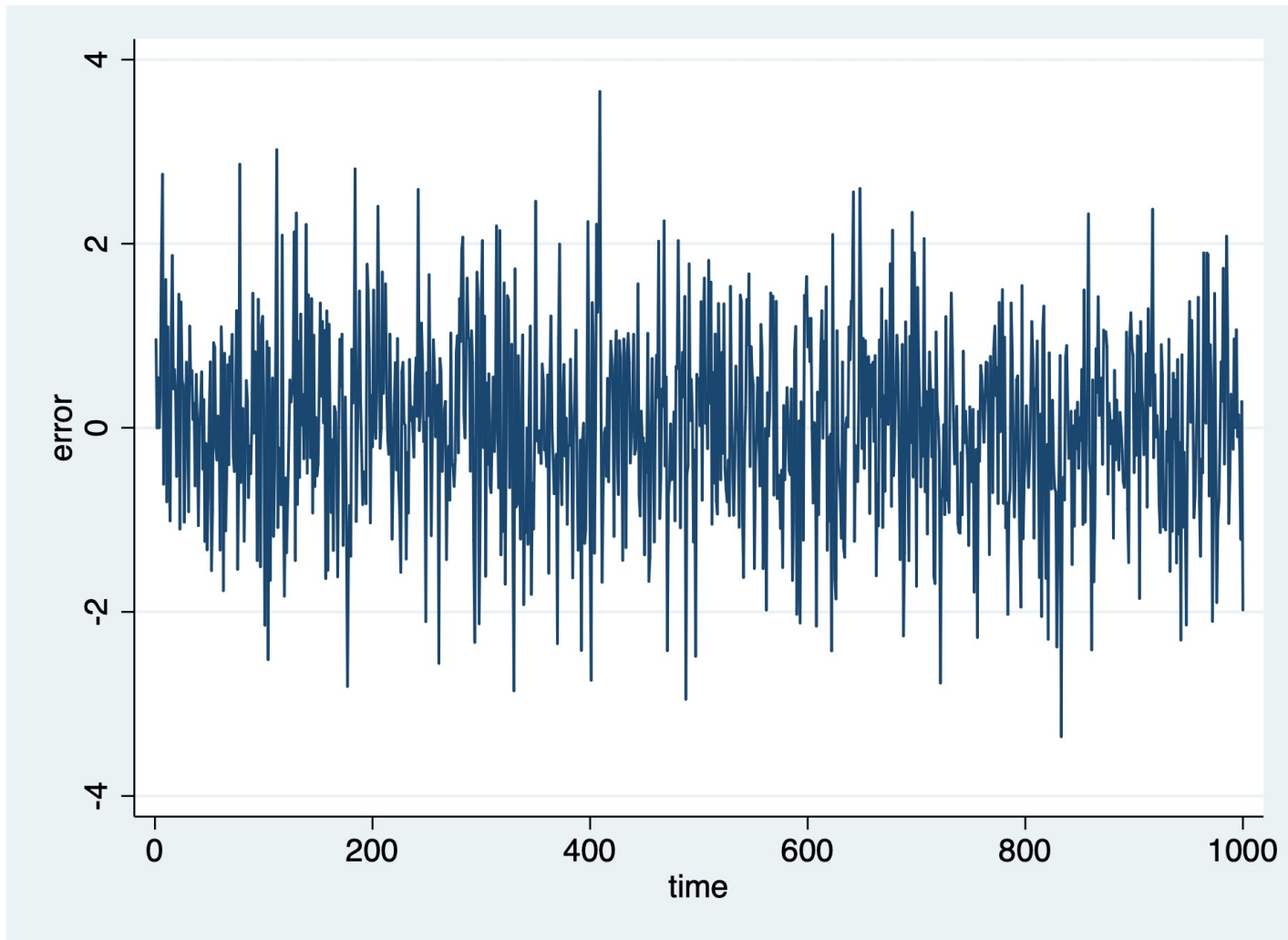
SIMPLE AND IMPORTANT PROCESS TO UNDERSTAND TIME SERIES MODELING: A WHITE NOISE PROCESS

- A White Noise process is a stochastic process where all observations are independently, identically distributed with zero mean, constant and finite variance and covariances equal to zero for any $t \neq s$.
- Example:

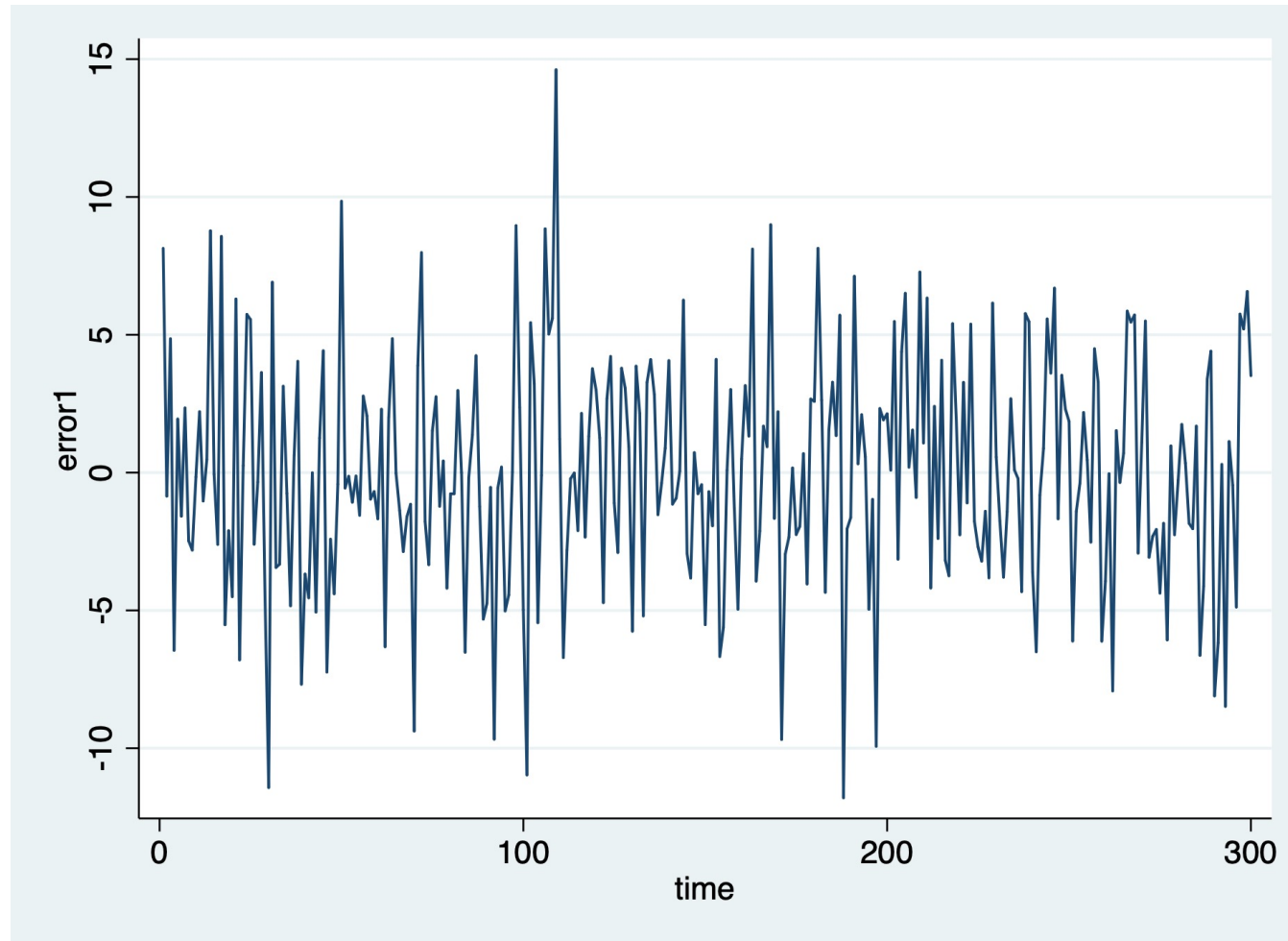
$$X_t = e_t, e_t \sim N(0,1)$$

.Show example of a simulated white noise process.

White Noise process $e_t \sim N(0,1)$



White Noise process $e_t \sim N(0,4)$



ARIMA(P,D,Q) PROCESSES

- ARIMA MODELS WERE SIMPLE MODELS THAT OUTPERFORMED THE LARGE SCALE SYSTEMS-OF-EQUATIONS APPROACH TO MACROECONOMIC FORECASTING BY COWLES COMMISSION.
- **COMPONENTS OF THE ARIMA MODELS:**
 - AUTOREGRESSIVE (AR)
 - MOVING AVERAGE (MA)
 - INTEGRATED (I)

Stationary AR(1) Models

Let X_t be an AR(1) process:

$$X_t = \beta X_{t-1} + e_t$$

- The linear dependence between X_t and X_{t-1} is given by correlation coefficient $\rho_{t,t-1} = \rho_1$ or so-called autocorrelation coefficient.
- For this process to be stationary (we will prove in chapter 4) we need $|\beta| < 1$.
- Using lag-operator, we can write the AR(1) model as:

$$(1 - \beta L)X_t = e_t$$

The mean, variance and covariance of the AR(1) process

$$X_t = \beta X_{t-1} + e_t$$

$$X_t = \beta^t X_0 + \sum_{i=1}^t \beta^{t-i} e_i$$

The plim t (tends to infinite) is $X_t = \sum_{i=1}^{\infty} \beta^{t-i} e_i$ when $|\beta| < 1$

Computing the Expected value of X_t :

$$E[X_t] = 0$$

Important: X_t is said to be stationary if it can be represented as a convergent infinite order moving average process

The mean, variance and covariance of the AR(1) process

$$\text{Var}(X_t) = E(\beta X_{t-1} + e_t)^2 = E(\beta^2 X_{t-1}^2 + 2e_t \beta X_{t-1} + e_t^2)$$

$$\text{Var}(X_t) = \beta^2 \text{Var}(X_t) + \sigma_e^2$$

$$\text{Var}(X_t) = \frac{\sigma_e^2}{1 - \beta^2}$$

$$\text{Cov}(X_t, X_{t-1}) = E(X_t X_{t-1}) - E(X_t)E(X_{t-1})$$

$$\text{Cov}(X_t, X_{t-1}) = \beta E(X_t^2)$$

$$\text{Cov}(X_t, X_{t-1}) = \beta \frac{\sigma_e^2}{1 - \beta^2}$$

Problems with OLS in Models with Lag-dependent variables

- Estimation: The OLS estimator are biased (although consistent) due the lagged dependent variable. We estimate by MLE. If the errors are autocorrelated, then the OLS estimator are also inconsistent. (show the simulated example on page 18 and 19).

$$X_t = \beta X_{t-1} + e_t$$

$$\hat{\beta}_{OLS} = \frac{Cov(X_t, X_{t-1})}{Var(X_{t-1})} = \frac{\sum X_t X_{t-1}}{\sum X_{t-1}^2}$$

$$\hat{\beta}_{OLS} = \frac{\sum (\beta X_{t-1} + e_t) X_{t-1}}{\sum X_{t-1}^2}$$

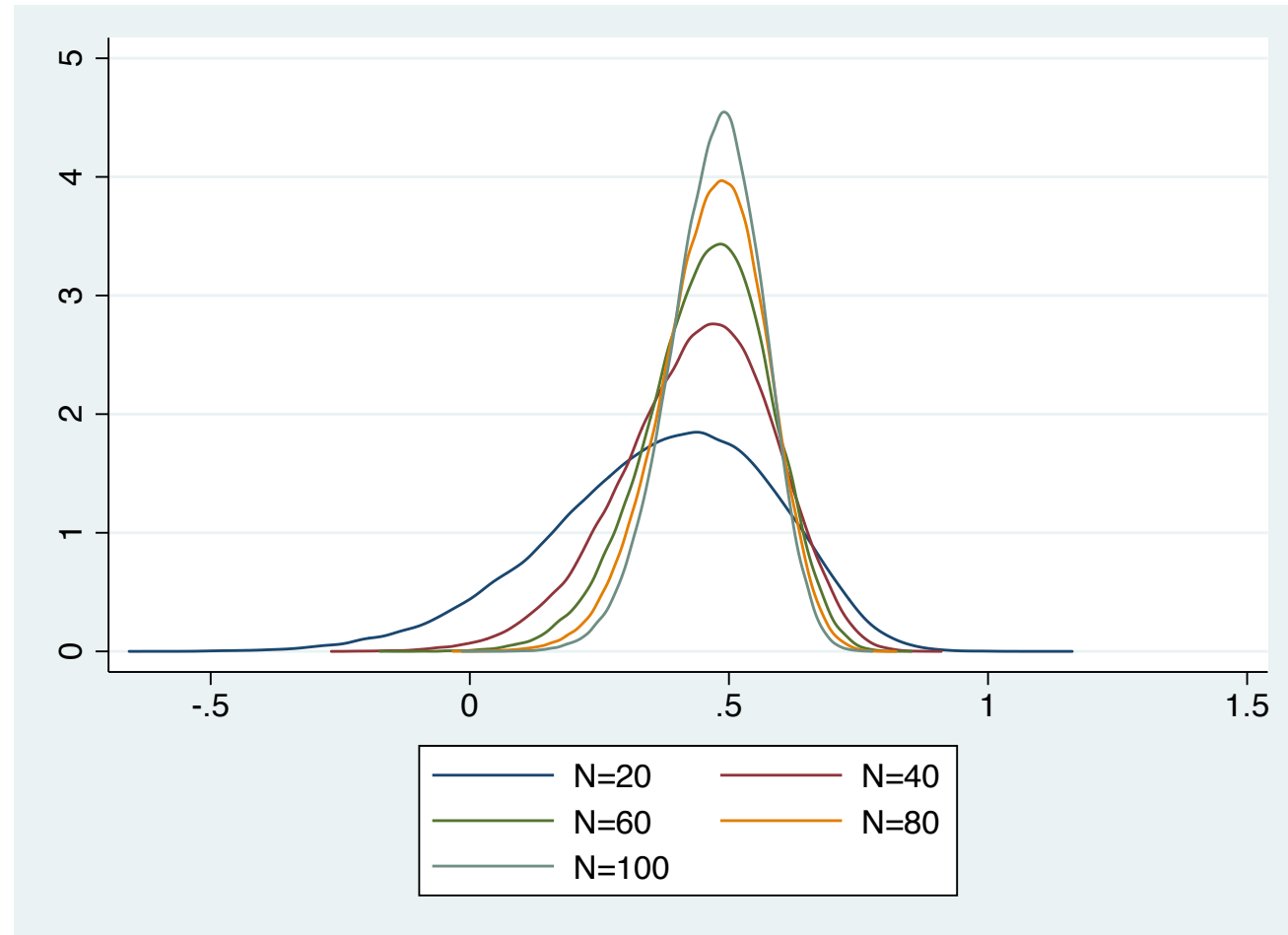
$$\hat{\beta}_{OLS} = \beta + \frac{\sum e_t X_{t-1}}{\sum X_{t-1}^2}$$

Problems with OLS in Models with Lag-dependent variables

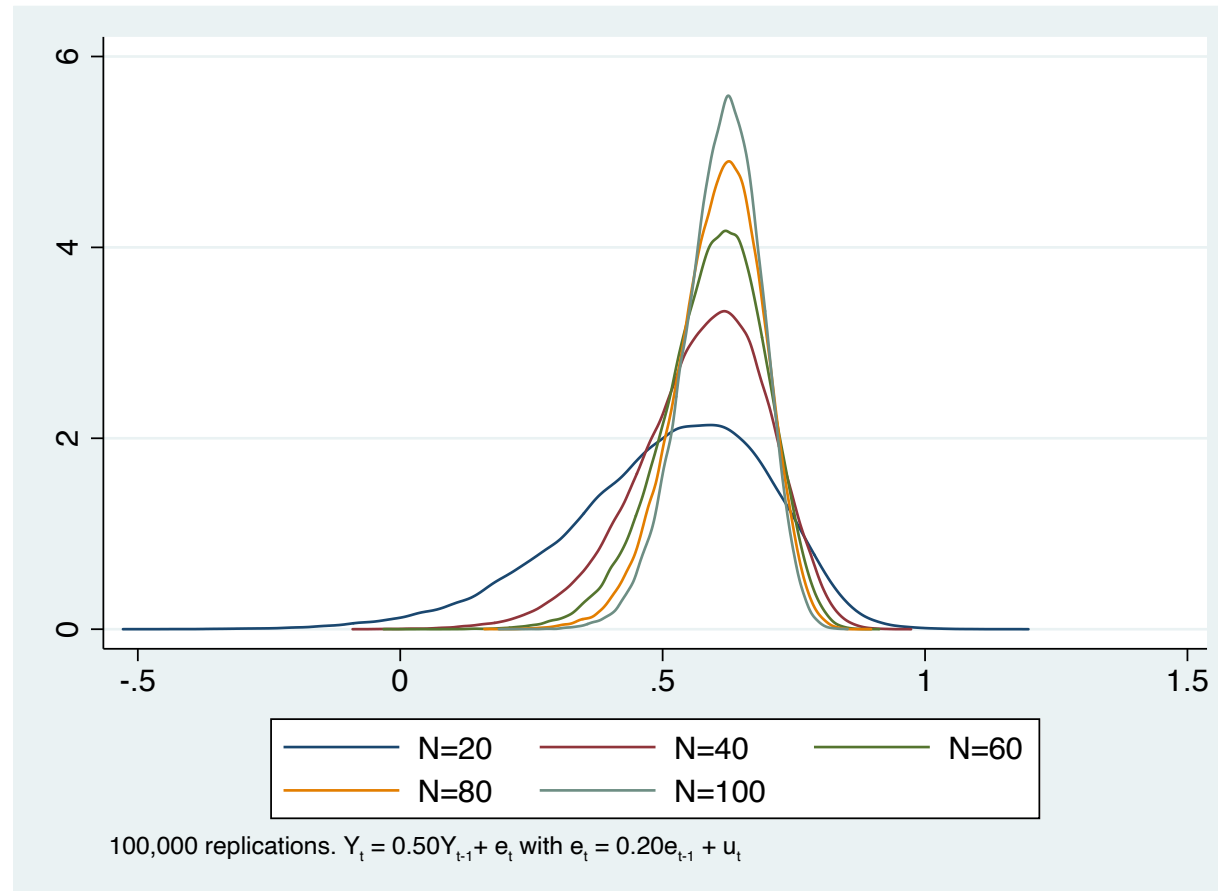
$$E(\hat{\beta}_{OLS}) = \beta + \frac{\sum E(e_t X_{t-1})}{\sum X_{t-1}^2}$$

- $E(e_t X_{t-1})$: is not equal zero. Since e_t directly affects X_t

OLS is biased but consistent ($X_t = 0.5X_{t-1} + e_t$)



OLS is biased and inconsistent
($X_t = 0.5X_{t-1} + e_t$; and $e_t = 0.2e_{t-1} + u_t$)



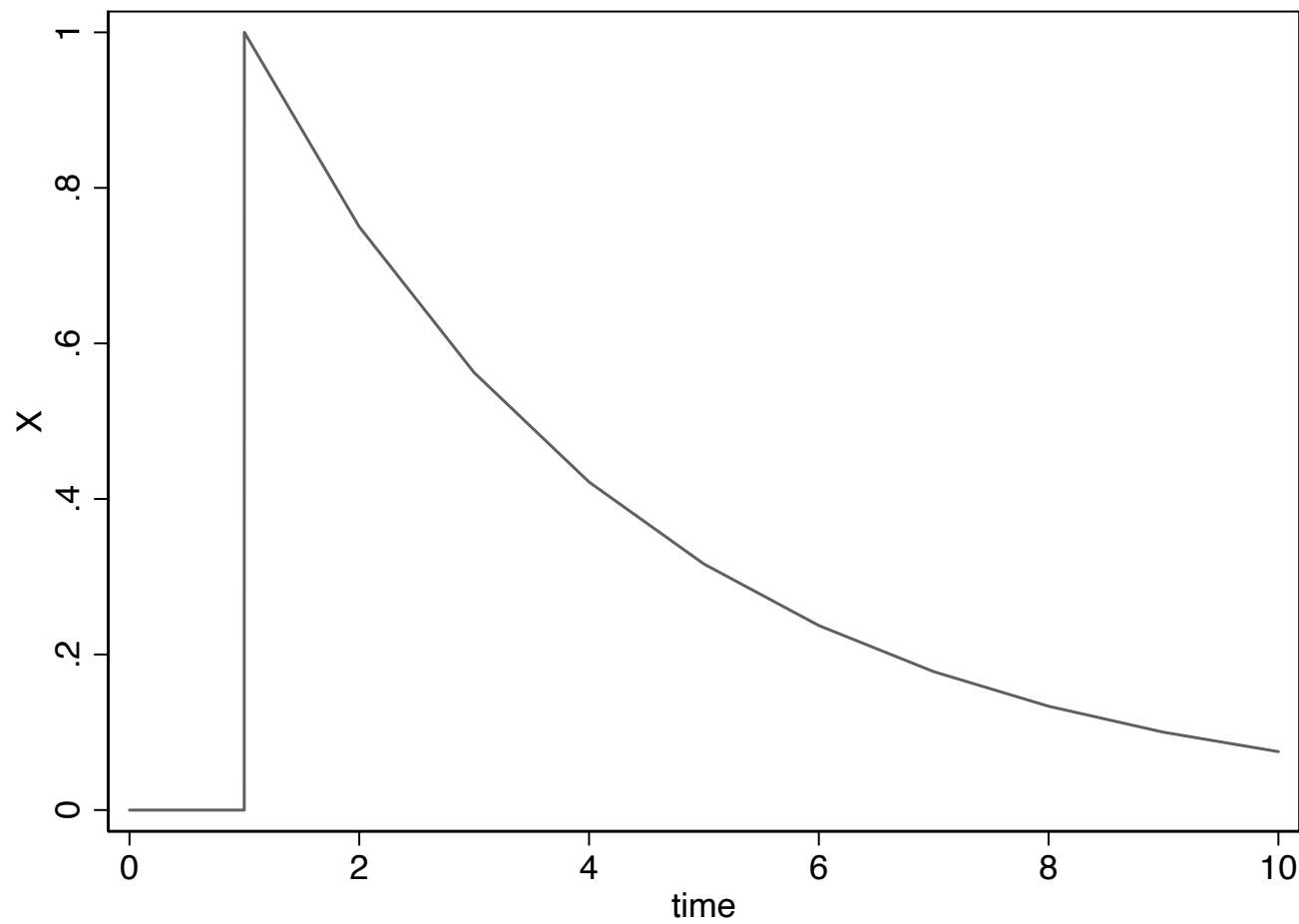
IMPULSE RESPONSES FUNCTION (IRF)

- How does a time series respond to an impulse shock (a particular shock on the error term)?
- Are shocks persistent or transitory?
- Example: AR(1) model: $X_t = 0.75X_{t-1} + e_t$
- Suppose the shock occurs on period t . Also, suppose that $X_t = 0$ until this moment, then in period t :

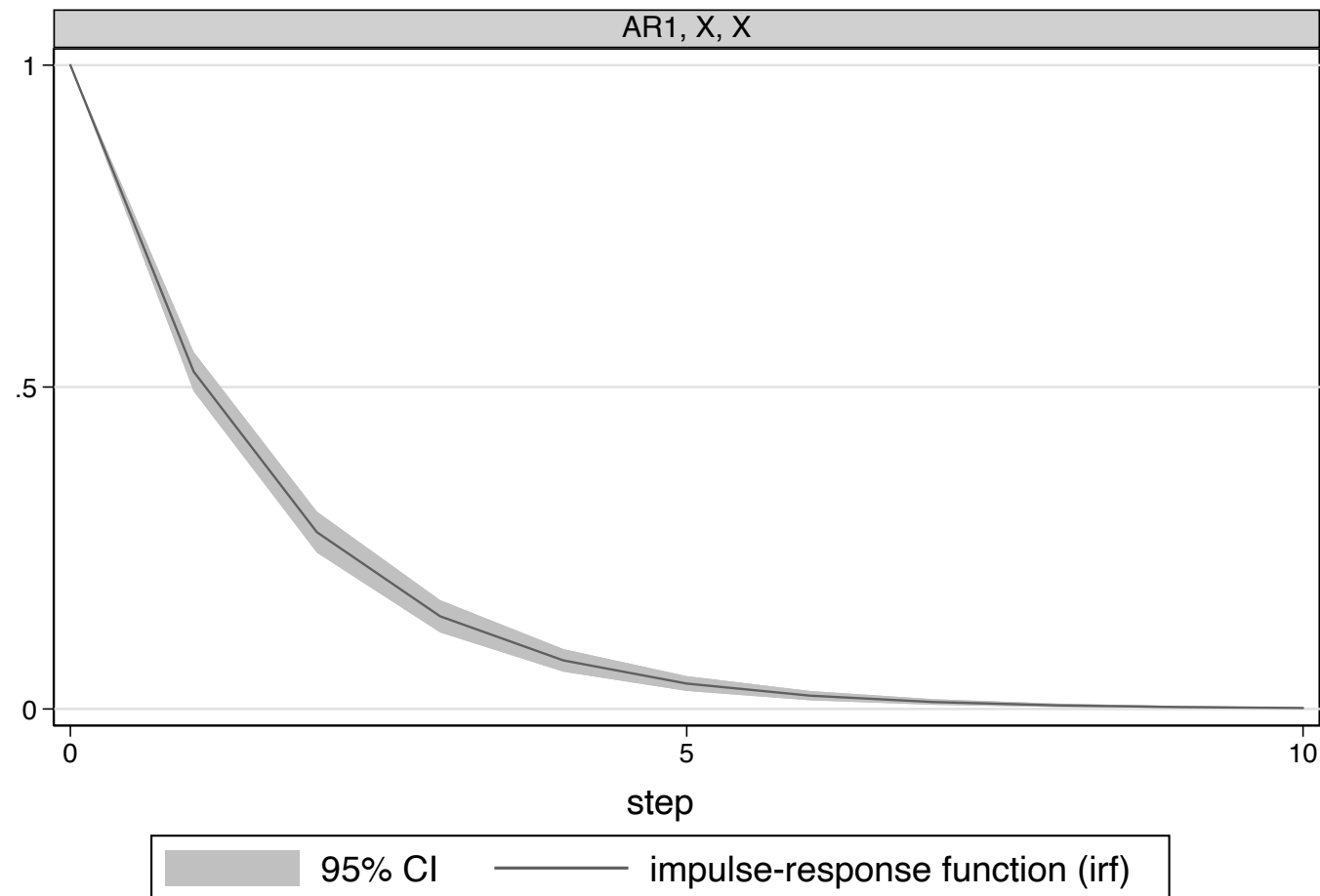
$$\begin{aligned}X_t &= 0.75(0) + 1 = 1 \\X_{t+1} &= 0.75(1) + 0 = 0.75 \\X_{t+2} &= 0.75(0.75) + 0 = (0.75)^2 \\&\text{and so on}\end{aligned}$$

- The IRF k periods after the shock is equal to β^k . Farther from X_0 , the effect of X_0 diminished
- The shock has an exponentially decaying effect of X (because $|\beta| < 1$)

Theoretical IRF: $X_t = 0.75X_{t-1} + e_t$



Estimated IRF AR(1): $X_t = 0.52404X_{t-1} + e_t$



Graphs by irfname, impulse variable, and response variable

AR(1): Forecasting

- Back to our previous example: AR(1) model: $X_t = 0.75X_{t-1} + e_t$
- Suppose that: $x_0 = 100$ and $e_1 = 20, e_2 = -30, e_3 = 10, e_4 = 15$
- $e_t \sim^{iid} N(0,100)$, we can compute the first four values of X:

$$X_1 = 0.75(100) + 20 = 95$$

$$X_4 = 0.75(40.9375) + 15 = 45.7031$$

We want to forecast X_5 , knowing up to X_4 .

We will compute the expected value of X_5 , conditional to all previous information:

$$E(X_5|X_4, X_3, \dots) = E(0.75X_4 + e_5|X_4, X_3, \dots) = 0.75X_4 + E(e_5|X_4, X_3, \dots) = 0.75(40.9375) + 0 = 30.7031$$

$$E(X_{t+a}|X_t, X_{t-1}, \dots) = \beta^a X_t$$

Stationary AR(p) Models

$$X_t = \beta_1 X_{t-1} + \beta_2 X_{t-2} + \cdots + \beta_p X_{t-p} + e_t$$

$$(1 - \beta_1 L - \beta_2 L^2, \dots, -L^p \beta_p) X_t = e_t$$

(Stationarity condition: all roots of the polynomial in L are outside the unit circle).

Example: AR(2) – Paul Samuelson's multiplier-accelerator model (closed economy with no government expenditure):

From the accounting identity:

$$Y_t = C_t + I_t$$

AR(2): an example

– Paul Samuelson's multiplier-accelerator model (closed economy with no government expenditure):

From the accounting identity:

$$Y_t = C_t + I_t$$

The Keynesian consumption function ($0 < \beta_1 = \text{MPC} < 1$):

$$C_t = \beta_1 Y_{t-1}$$

Also: $I_t = \beta_2 (C_t - C_{t-1}) + e_t$; substituting into the identity, we have:

$Y_t = \beta_1 Y_{t-1} + \beta_2 (\beta_1 Y_{t-1} - \beta_1 Y_{t-2}) + e_t = \beta_1 (1 + \beta_2) Y_{t-1} - \beta_1 \beta_2 Y_{t-2} + e_t$ is an AR(2) model.

AR(2): an example

$$Y_t = \beta_1(1 + \beta_2)Y_{t-1} - \beta_1\beta_2Y_{t-2}$$

Let: $a_1 = \beta_1(1 + \beta_2)$

$$a_2 = -\beta_1\beta_2$$

$$Y_t = a_1Y_{t-1} + a_2Y_{t-2}$$

Note that: $a_1 + a_2 = \beta_1$ (the MPC)

and $a_2 < 0$

$\Rightarrow$ this model implies a stationary solution.

AR(2): an example

- Blanchard(1981) estimated the AR(2) model using quarterly US data on detrended log GNP (y_t^d) for the period 47:3-78:4:
- $\widehat{y}_t^d = 1.34y_{t-1}^d - 0.42y_{t-2}^d$
- Note that: $\widehat{a}_1 + \widehat{a}_2 = 0.92 < 1$ and $\widehat{a}_2$ is negative, satisfying the conditions imposing by Samuleson's multiplier-accelerator model. Also, $\widehat{a}_1^2 + 4\widehat{a}_2 = 0.1156 > 0$ (no pseudo-cycles)
- Should we accept that this process is trend-stationary? $\widehat{a}_1 + \widehat{a}_2 = 0.92$. Is this value statistically different than 1? We will discuss later on unit root test and process with a near unit-root.

AR(p) Impulse Response Function

- As in the case of the AR(1), the effect of a shock will linger on, but the effects decay.

AR(p) Forecasting

- Let's have the following AR(3) model: $X_t = 0.75X_{t-1} + 0.5X_{t-2} + 0.10X_{t-3} + e_t$
- Suppose that: $X_1 = 5, X_2 = -10, X_3 = 15$ and $e_1 = 20, e_2 = -30, e_3 = 10, e_4 = 15$

$$E(X_4|X_3, X_2, \dots) = E(0.75X_{t-3} + 0.5X_{t-2} + 0.10X_{t-1} + e_4|X_3, X_2, \dots)$$

$$= 0.75(15) + 0.5(-10) + 0.10(5) + E(e_4|X_3, X_2, \dots) = 6.75$$

To forecast $E(X_5|X_3, X_4, \dots)$, we use the $E(X_4|X_3, X_2, \dots)$.

Moving Average Models: MA(1)

$$X_t = e_t$$

$$X_t = u_t + \beta u_{t-1}$$

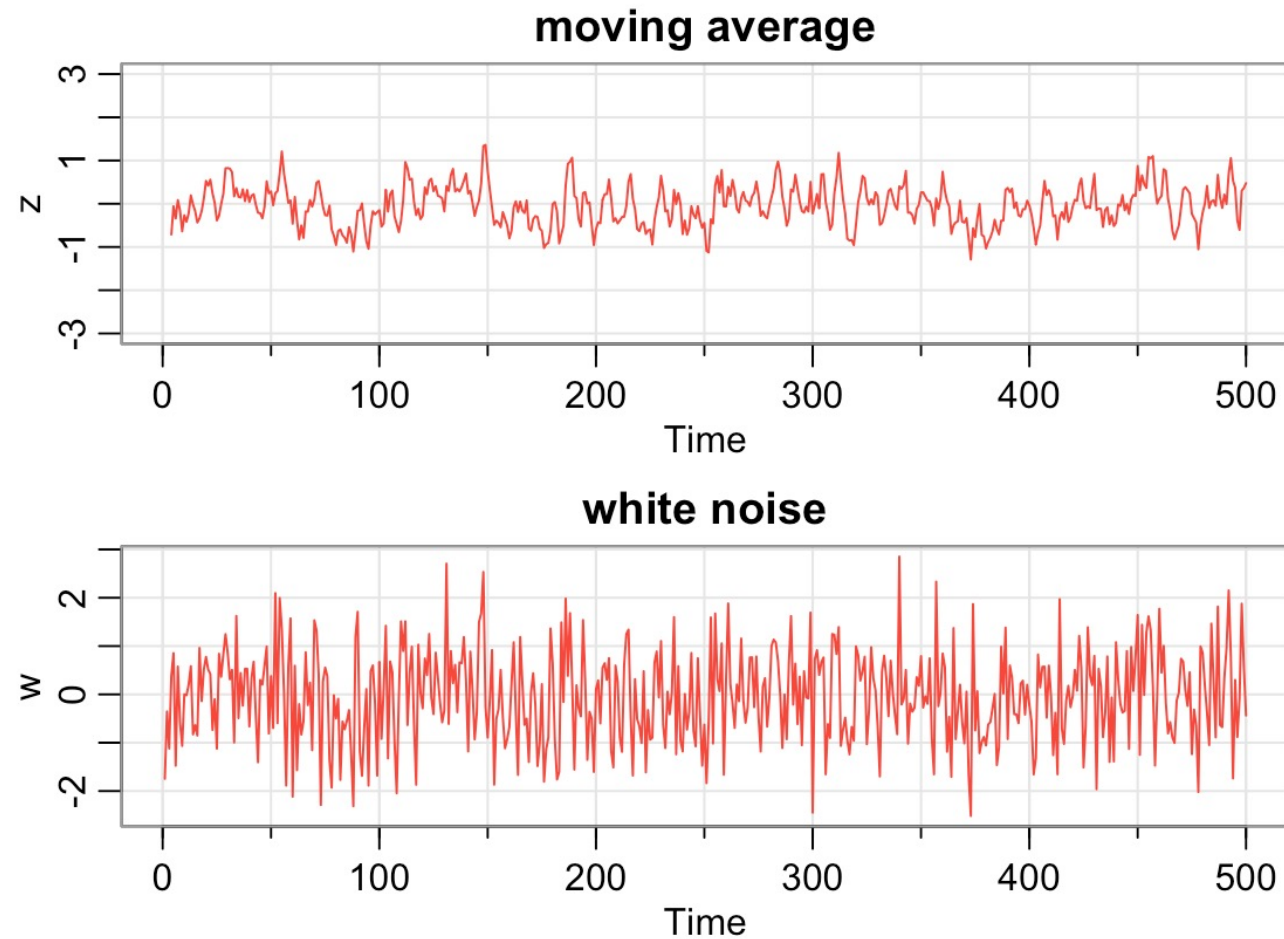
The errors e_t are autocorrelated, so X_t (we will see that and invertible MA(1) can be represented by and AR(∞), however (u_t are random shocks and $u_t \sim iidN(\mu, \sigma_u^2)$): not only u_t have a direct effect on X_t , but also the past of u_t .

Give Pierre's example about Canadian inflation (if we want to stabilize inflation, the coefficient of u_t has to be equal 1).

MA(q) process: an example

- Consider the following random experiment:
- Toss a coin, if head, you win \$1 and if tail you lose a \$1.
- Denote the outcome of your experiment by u_t (note that u_t is iid, with mean equal zero and constant variance):
 - $$\begin{cases} u_t = 1 & \text{if head} \\ u_t = -1 & \text{if tail} \end{cases}$$
- Denote the random variable X_t as the average payoff on the last four tosses:
- $X_t = \frac{1}{4}u_t + \frac{1}{4}u_{t-1} + \frac{1}{4}u_{t-2} + \frac{1}{4}u_{t-3}$ is an MA(3)

Moving-Average Process: show smaller variance than the white noise (due the average)



MA(1) Impulse Response Function

- Assuming X follows the following invertible MA(1) process:

$$X_t = u_t + 0.75u_{t-1}$$

- For this process to be invertible (we will prove in chapter 4) we need $|\beta| < 1$.
- Suppose the shock occurs on period $t = 1$. Also, supposed that $X_t = 0$ and $e_t = 0$ for every period this period. At $t = 1$ X_1 receive a one-time shock equal to 1 ($u_1 = 1, u_2 = 0, u_3 = 0$)

$$X_1 = u_1 + 0.75u_0 = 1$$

$$X_2 = u_2 + 0.75u_1 = 0.75$$

$$X_3 = u_3 + 0.75u_2 = 0$$

$$X_4 = u_4 + 0.75u_3 = 0$$

- In a MA process, memories are short and the economy recovers quickly.

Forecasting and Moving Average Model

- First estimate the model and calculate the residuals (they will be the estimated errors); we use the residuals to forecast X.
- Let's have the following estimated MA(1):

$$\widehat{X}_t = r_t + \hat{\beta}r_{t-1}$$

$$E(X_1|r_0) = E(r_1) + 0.5120095(r_0) = 0.5120095r_0$$

$$E(X_t|r_{t-1}) = 0.5120095r_{t-1}$$

Forecasting and Moving Average Model: STATA Commands

use MAexamples.dta (if you are using the series generated, then you do not need to generate time)

gen time = _n

tsset time

arima X, ma(1) nocons nolog (this is how to estimate the MA model on STATA)

predict Xhat

predict r, residuals

list X Xhat r in 2996/3000

tsappend, add(1)

drop Xhat

predict Xhat (to recalculate the fitted values)

list X Xhat r in 2996/3001

MA(q) Impulse Response Function (IRF)

$$X_t = u_t + \beta_1 u_{t-1} + \beta_2 u_{t-2} + \dots + \beta_q u_{t-q}$$

MA(q) IRF: Suppose the shock occurs on period $t = k$. Also, supposed that $X_t = 0$ and $e_t = 0$ for every period this period. At $t = k$ X_k receives a one-time shock equal to $(u_k = 1, u_{k+1} = 0, u_{k+2} = 0, u_{k+3} = 0)$:

$$X_k = 1 + \beta_1(0) + \beta_2(0) + \dots + \beta_q(0) = 1$$

$$X_{k+1} = 0 + \beta_1(1) + \beta_2(0) + \dots + \beta_q(0) = \beta_1$$

$$X_{k+q} = 0 + \beta_1(0) + \beta_2(0) + \dots + \beta_q(1) = \beta_q$$

- The IRF of and MA(q) is only non-zero for q periods ahead.

Non-zero mean stationary AR(p) Models

Let's X_t be an AR(1) process:

$$X_t = \beta_0 + \beta_1 X_{t-1} + e_t$$

- A stationary process will have: $E(X_t) = E(X_{t-1})$, then:

$$E(X_t) = \beta_0 + \beta_1 E(X_t)$$

$$E(X_t) - \beta_1 E(X_t) = \beta_0$$

$$E(X_t) = \frac{\beta_0}{1 - \beta_1}$$

Recall that for this process to be stationary (we need $|\beta| < 1$).

Non-zero mean stationary AR(p) Models

- A stationary process will have: $\text{Var}(X_t) = \text{Var}(X_{t-1})$, then:

Then, the variance of the AR(1) process will be:

$$\text{Var}(X_t) = \beta_1^2 \text{Var}(X_t) + \text{Var}(e_t)$$

$$\text{Var}(X_t) - \beta_1^2 \text{Var}(X_t) = \text{Var}(e_t)$$

$$\text{Var}(X_t) = \frac{\sigma_e^2}{1-\beta_1^2}$$

(the constant β_0 does not affect the variance)

Non-zero mean stationary AR(p) Models

- Now, let's X_t be an AR(p) process:

$$X_t = \beta_1 X_{t-1} + \beta_2 X_{t-2} + \dots + \beta_p X_{t-p} + e_t$$

Then, the mean of this process will be:

$$E(X_t) = \beta_0 + \beta_1 E(X_t) + \beta_2 E(X_t) + \dots + \beta_q E(X_t)$$

$$E(X_t) = \frac{\beta_0}{1 - \beta_1 - \beta_2 - \dots - \beta_q}$$

OBS: when we estimate the model we can include the constant or estimate the de-meaning variable (see example page 43 Levendis).

Non-zero mean stationary MA(q) Models

Let's X_t be a MA(1) process:

$$X_t = \alpha + u_t + \beta u_{t-1}$$

$$u_t \sim iidN(0, \sigma_u^2)$$

$$E(X_t) = \alpha + E(u_t) + \beta E(u_{t-1})$$

$$E(X_t) = \alpha$$

The mean of an MA(q) also will be equal to the constant term.

Non-zero mean stationary MA(q) Models

The variance of a MA(1) process:

$$X_t = \alpha + u_t + \beta u_{t-1}$$

$$u_t \sim iid N(0, \sigma_u^2)$$

$$Cov(u_t, u_t) = Var(u_t) = \sigma_u^2$$

$$Cov(u_t, u_{t-j}) = E(u_t \cdot u_{t-j}) = 0, \quad \text{for } t \neq j$$

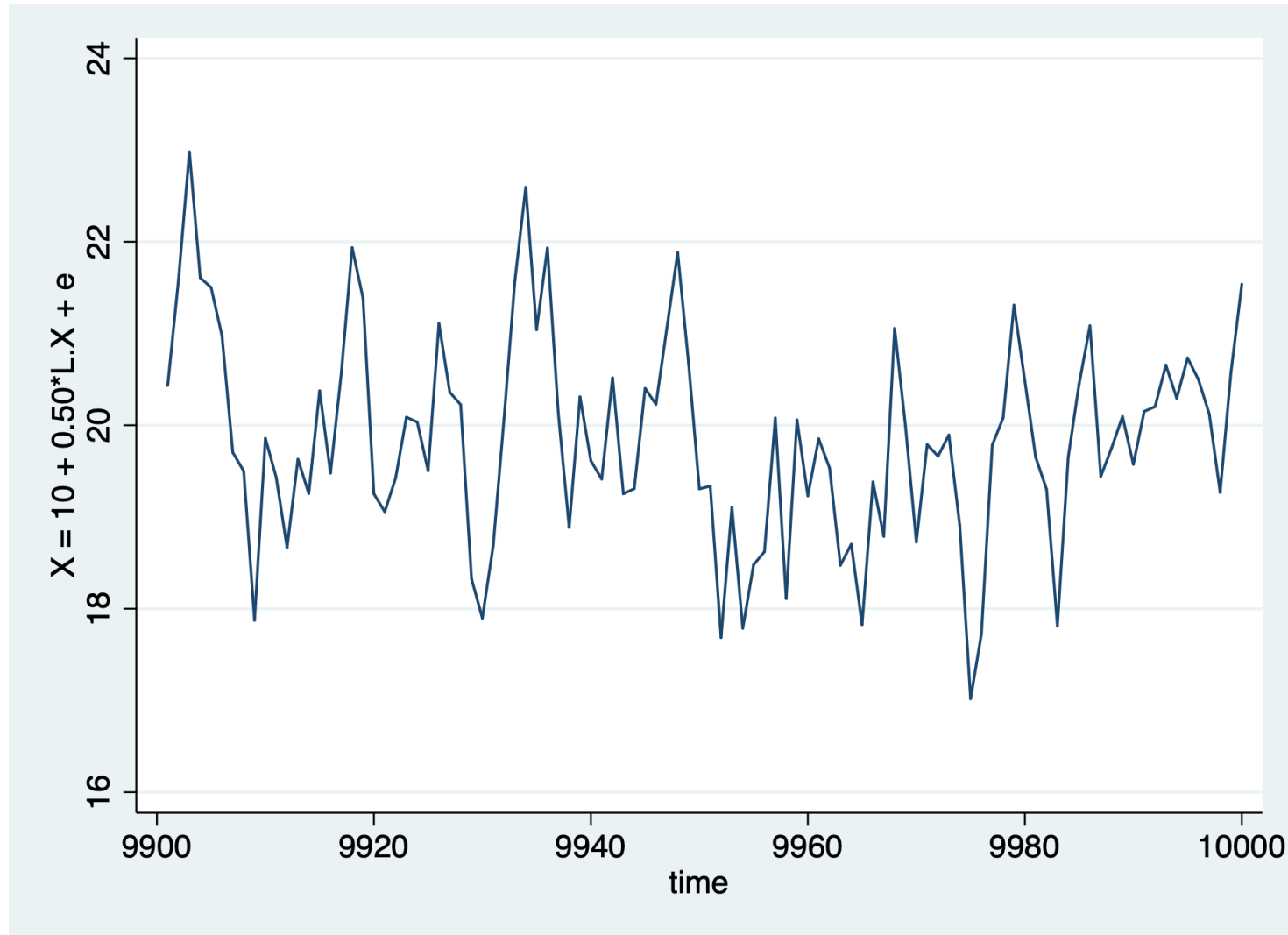
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Simulating a non-zero mean stationary AR(1) process

```
set more off
drop _all
clear all
set obs 10000
set seed 1234
gen time = _n
tsset time
gen double error = rnormal()
label var e "errors"
gen double X = error in 1
replace X = 10 + 0.50*L.X + error in 2/L
drop if _n <= 1000
label var X "X = 10 + 0.50*L.X + e"
sum X
save "/Users/cati/Desktop/AR1nonzero.dta", replace
graph twoway line X time if time > 9900
```

Simulating a non-zero mean stationary AR(1) process:

$$X_t = 10 + 0.5X_{t-1} + e_t$$



AR(2) with pseudo-cycle (characteristics roots are complex)

$$X_t = 1.5X_{t-1} - 0.75X_{t-2} + e_t$$

